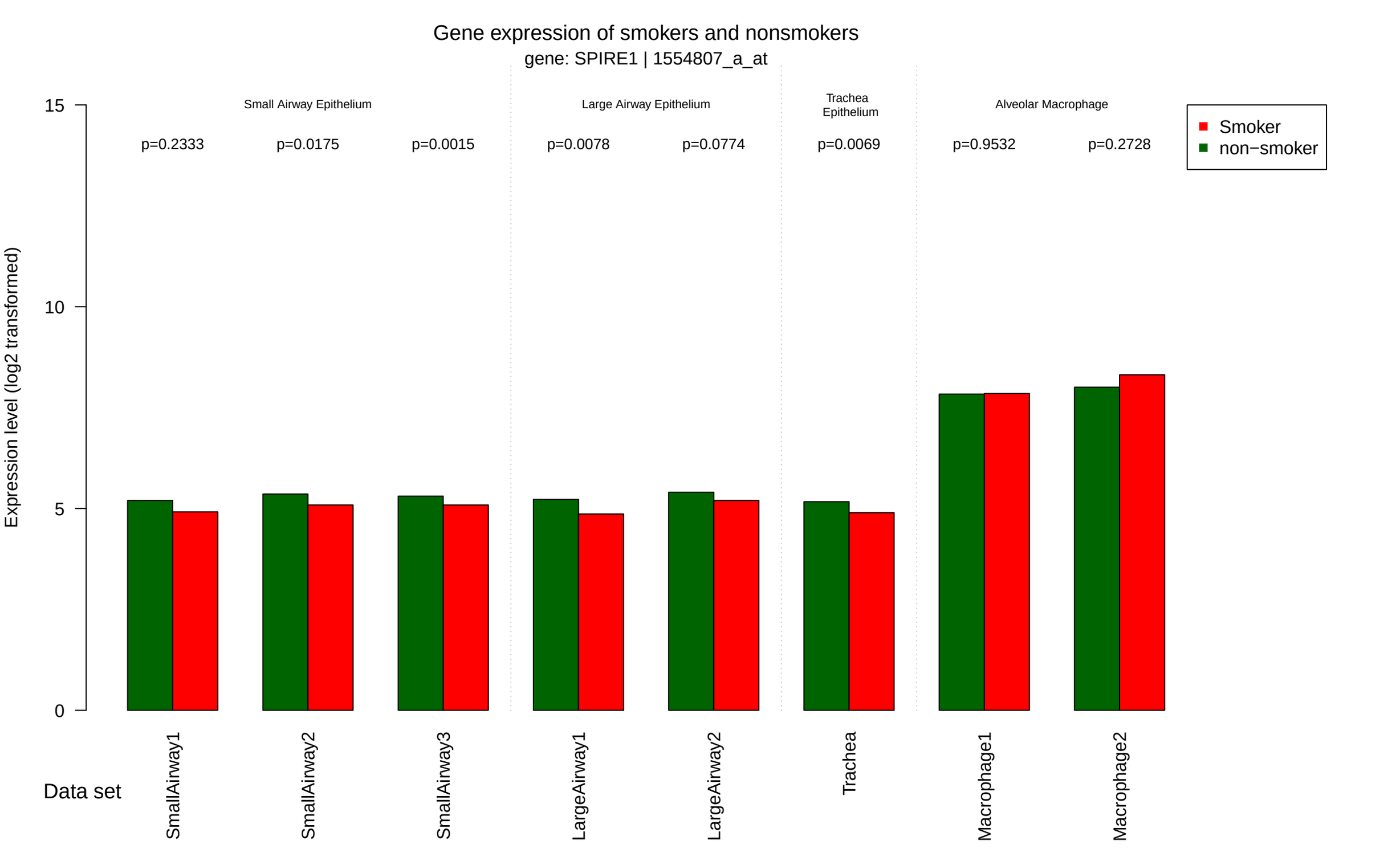
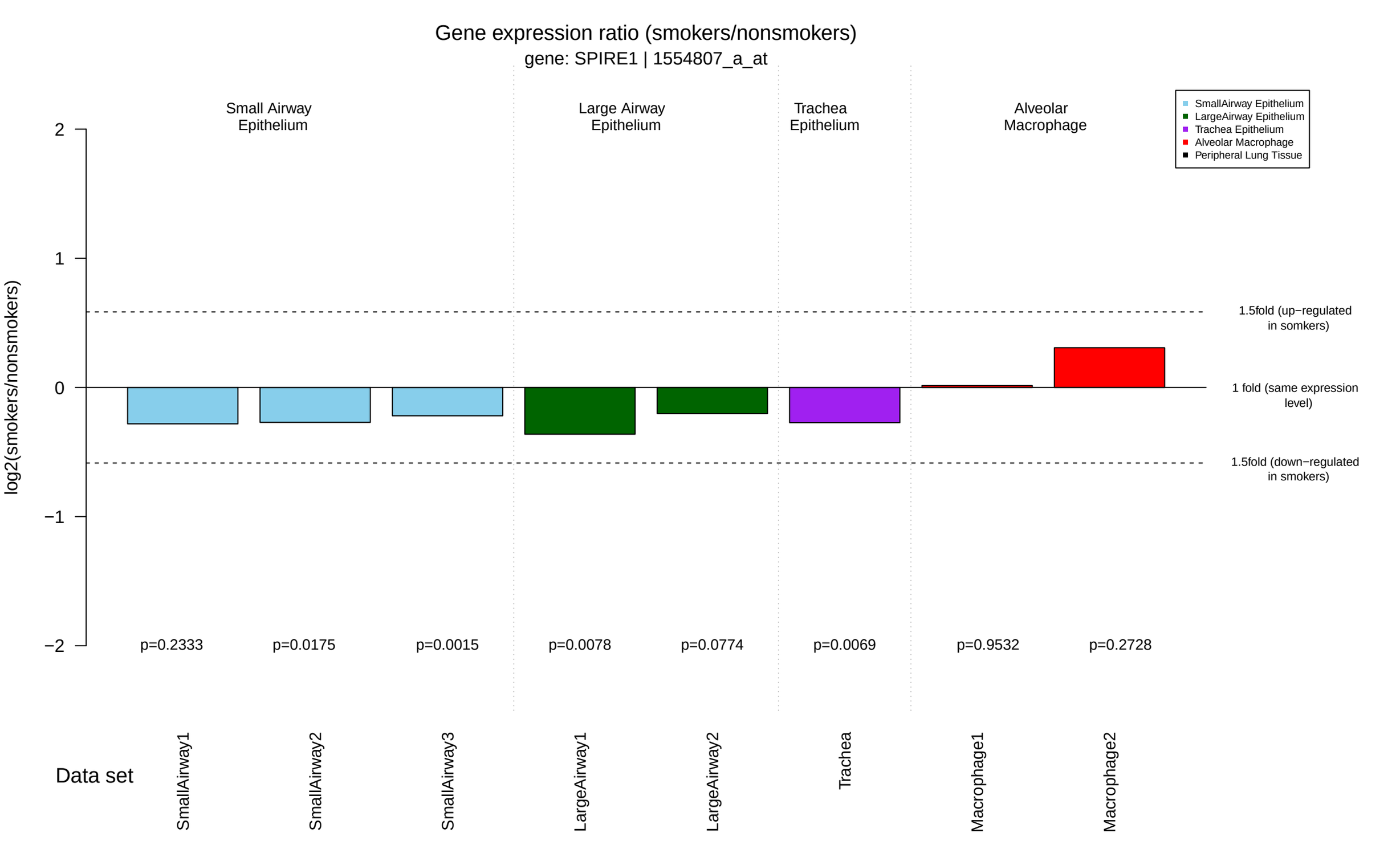
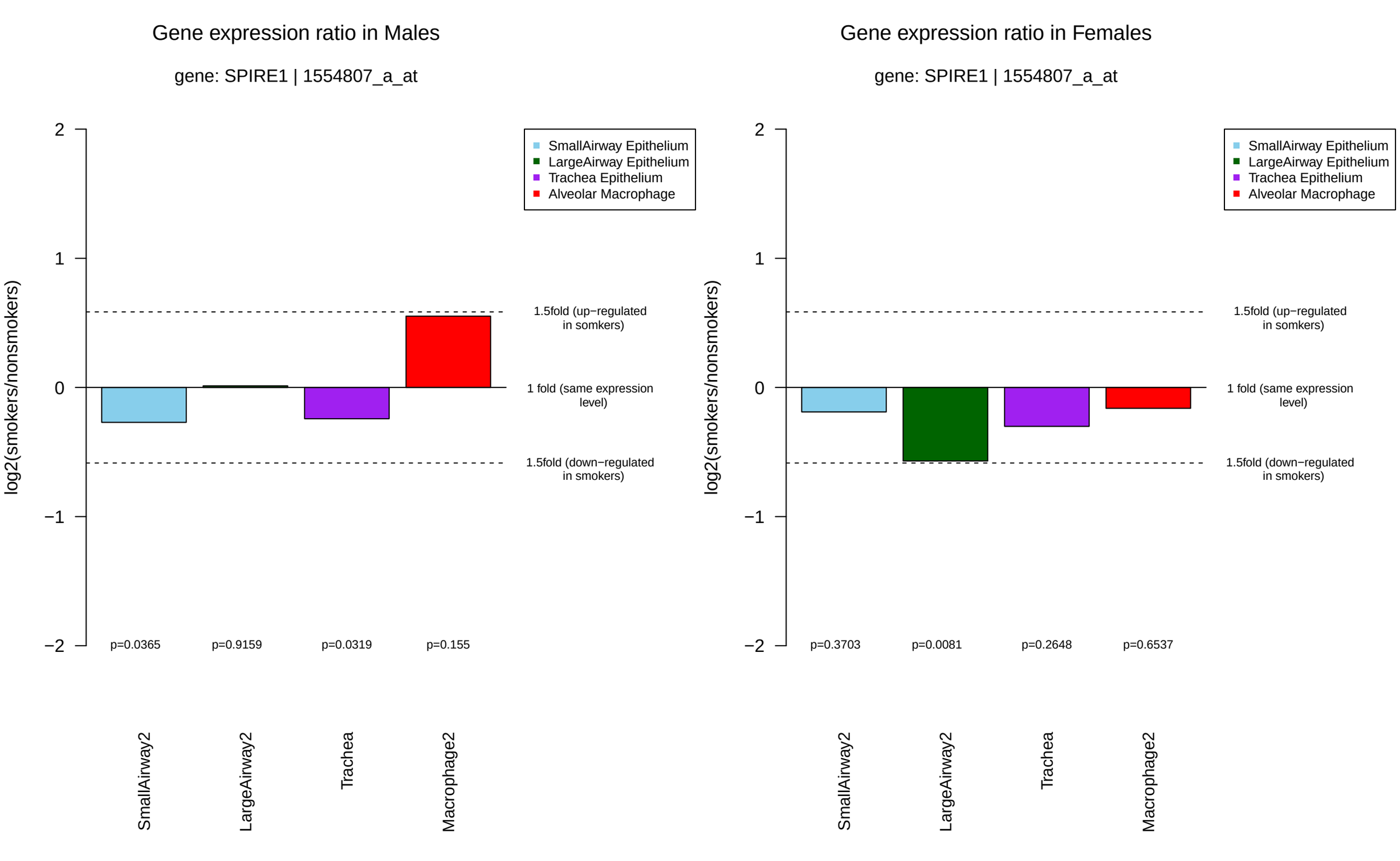
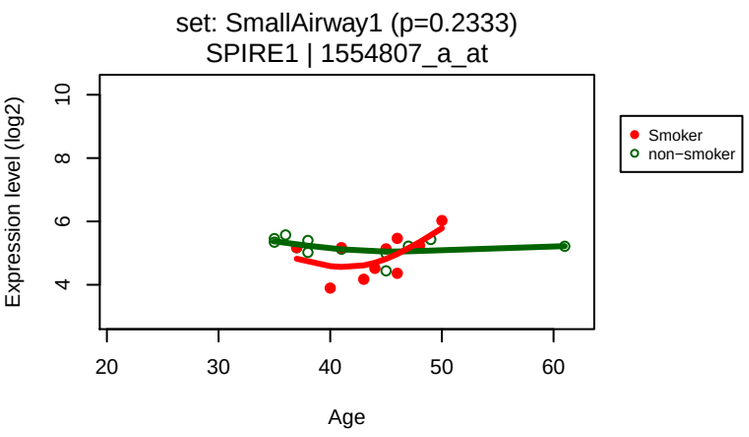
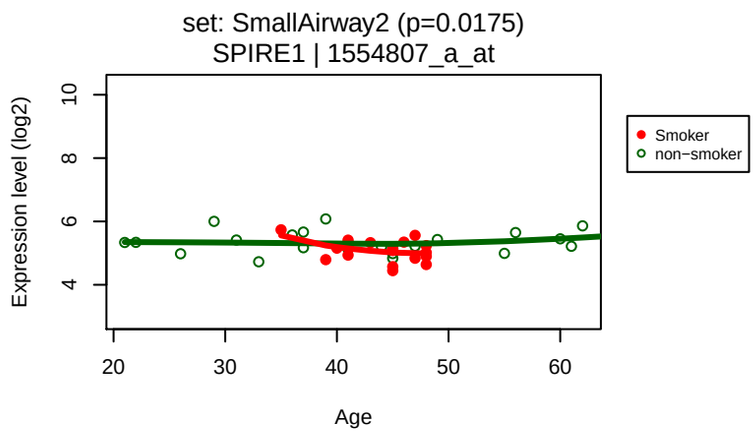
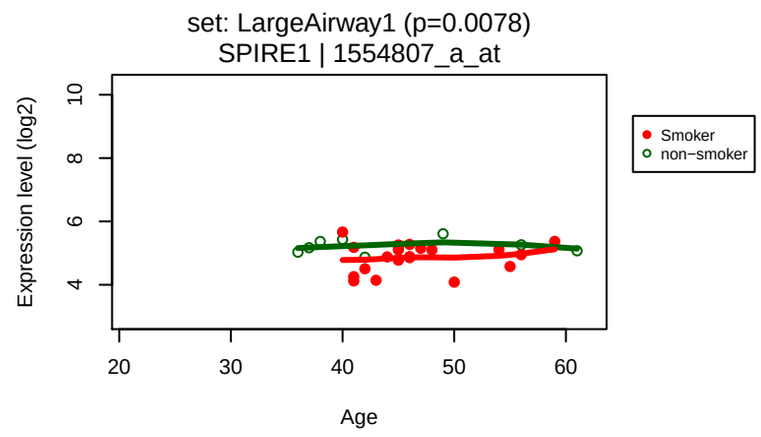

Gene expression of smokers and nonsmokers
gene: SPIRE1 | 1554807_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2333
p=0.0175
p=0.0015
p=0.0078
p=0.0774
p=0.0069
p=0.9532
p=0.2728
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

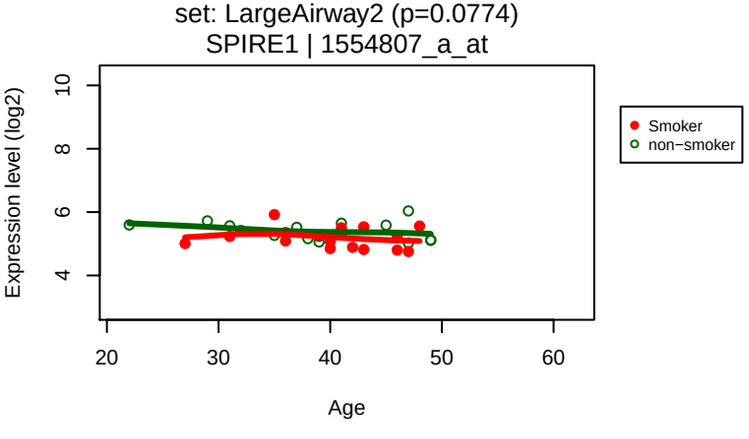

Gene expression ratio (smokers/nonsmokers)
gene: SPIRE1 | 1554807_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2333
p=0.0175
p=0.0015
p=0.0078
p=0.0774
p=0.0069
p=0.9532
p=0.2728
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

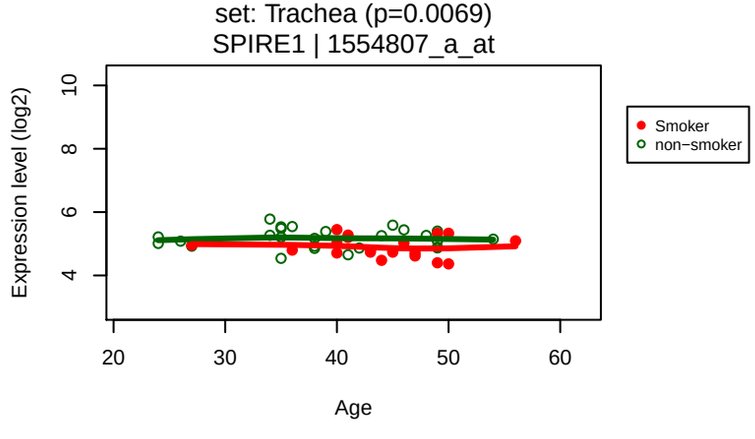

Gene expression ratio in Males
gene: SPIRE1 | 1554807_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0365
p=0.9159
p=0.0319
p=0.155
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPIRE1 | 1554807_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3703
p=0.0081
p=0.2648
p=0.6537
SmallAirway2
LargeAirway2
Trachea
Macrophage2

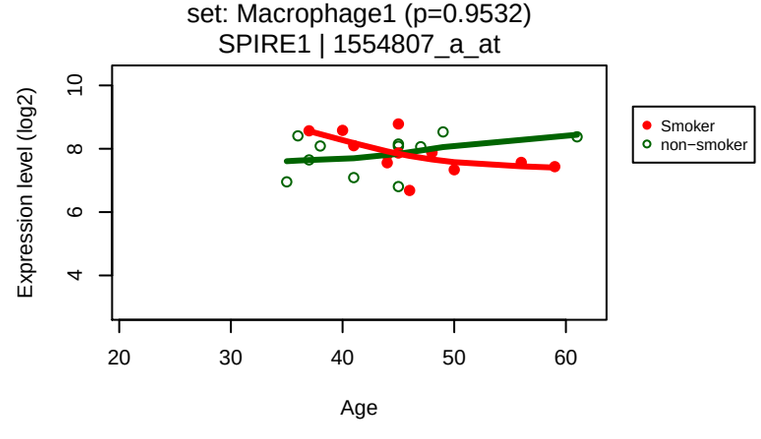

set: SmallAirway1 (p=0.2333)
SPIRE1 | 1554807_a_at
Expression level (log2)
Age
Smoker
non−smoker

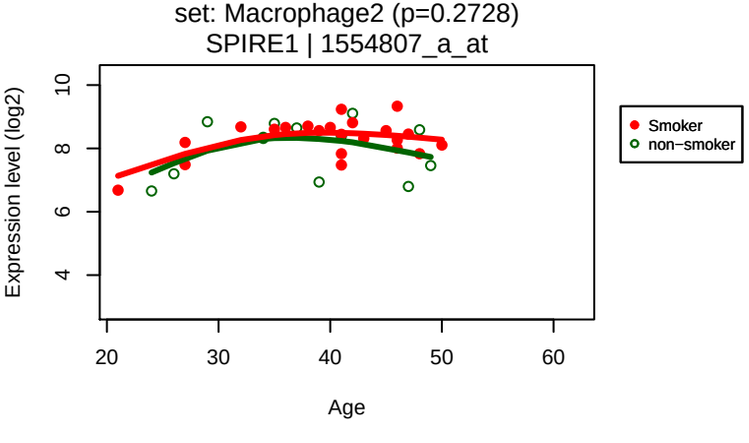

set: SmallAirway2 (p=0.0175)
SPIRE1 | 1554807_a_at
Expression level (log2)
Age
Smoker
non−smoker

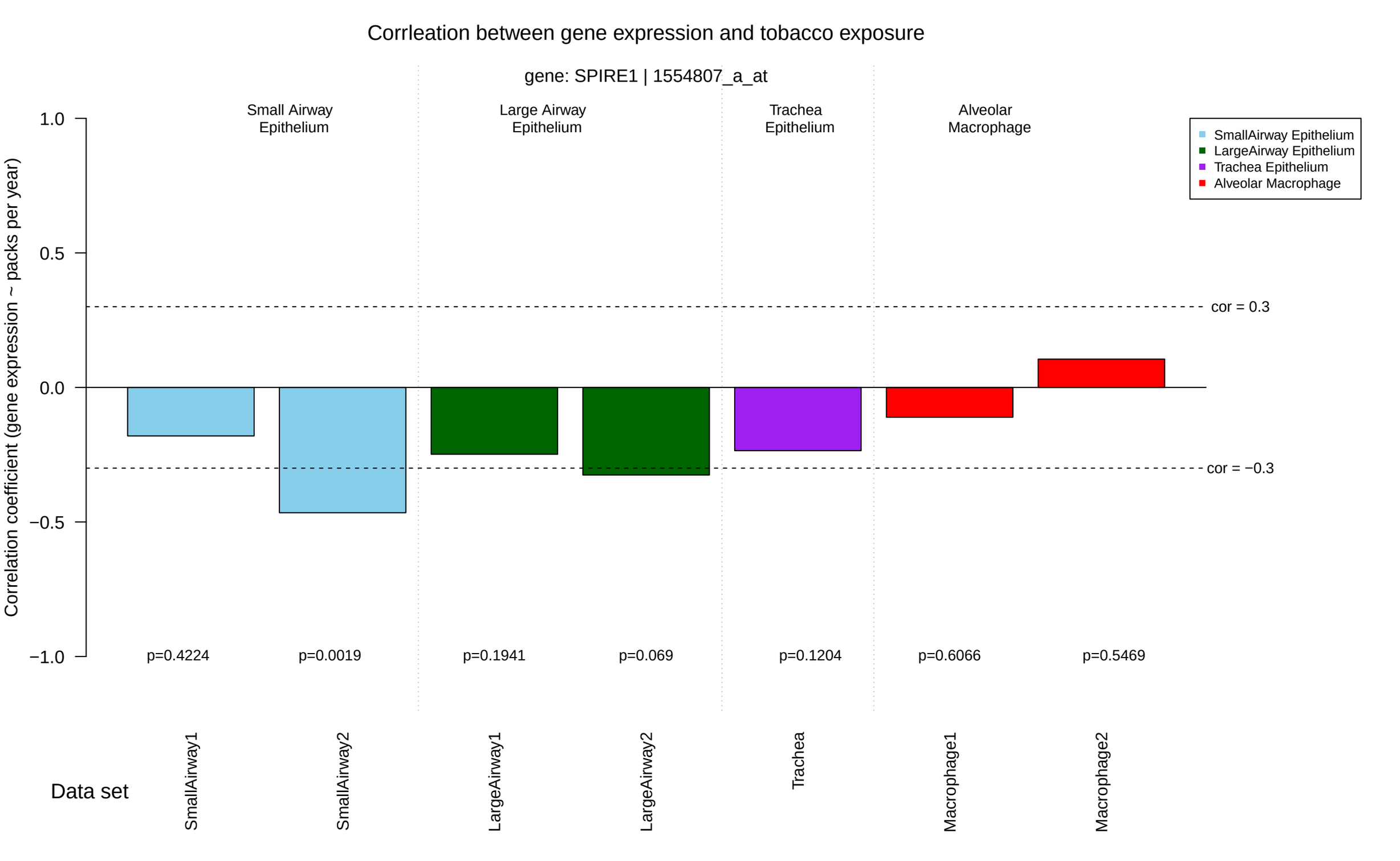

set: LargeAirway1 (p=0.0078)
SPIRE1 | 1554807_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0774)
SPIRE1 | 1554807_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0069)
SPIRE1 | 1554807_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9532)
SPIRE1 | 1554807_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2728)
SPIRE1 | 1554807_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPIRE1 | 1554807_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4224
p=0.0019
p=0.1941
p=0.069
p=0.1204
p=0.6066
p=0.5469
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2